

Lecture 23: Green's Theorem

Background & Motivation

Green's Theorem is the first of the three big theorems in vector calculus. It converts a line integral around a closed curve into a double integral over the enclosed region. This is powerful both computationally—sometimes one side is much easier than the other—and theoretically, as it connects circulation to curl.

1. Green's Theorem

– Converts a line integral around a closed curve into a double integral over the enclosed region (and vice versa)

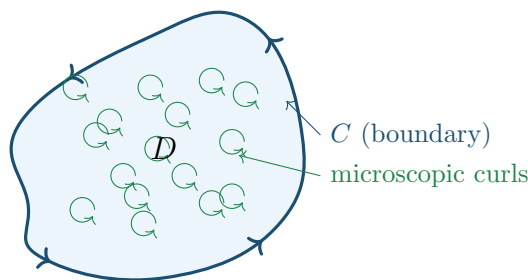
Theorem 1: Green's Theorem (Circulation Form)

Let C be a positively oriented (CCW), simple, closed curve bounding a region D . If P and Q have continuous partial derivatives on an open region containing D :

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Requirements:

- C must be **simple, closed, positively oriented**
- P, Q must be **continuously differentiable on all of D**



Macroscopic circulation around C = sum of all microscopic curls $(Q_x - P_y)$ inside D .

Theorem 2: Green's Theorem (Vector Form)

Recall that $\oint_C P dx + Q dy$ is really a line integral of $\mathbf{F} = \langle P, Q \rangle$ along C . Writing Green's Theorem

entirely in terms of $\mathbf{F}$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

The middle form is what we compute directly (parameterize C , dot, integrate). The right side is what Green's lets us swap to—a double integral over D .

Preview: The quantity $Q_x - P_y$ is what we will call the **curl** of $\mathbf{F}$ (next lecture). It measures local rotation. When we later learn **Stokes' Theorem**, we will see that Green's is just Stokes' with the surface flat in the xy -plane. Swap $Q_x - P_y$ for curl and you get Stokes'.

When to use Green's Theorem

- Hard line integral but easy double integral $\implies$ use Green's left-to-right
- Hard double integral but easy line integral $\implies$ use Green's right-to-left
- To compute areas via line integrals

Example 1: Verification on a Triangle (Both Sides)

Verify Green's Theorem for $\oint_C (x^2y dx + xy^2 dy)$ around the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$.

Double integral side:

1. **Compute:** $P = x^2y$, $Q = xy^2$. $Q_x - P_y = y^2 - x^2$
2. **Set up:** $\iint_D (y^2 - x^2) dA = \int_0^1 \int_0^{1-x} (y^2 - x^2) dy dx$
3. **Inner integral:** $\left[\frac{y^3}{3} - x^2y \right]_0^{1-x} = \frac{(1-x)^3}{3} - x^2(1-x)$
4. **Evaluate:** $\int_0^1 \left[\frac{(1-x)^3}{3} - x^2(1-x) \right] dx = \frac{1}{12} - \frac{1}{12} = 0$

Line integral side:

- C_1 : $(0,0) \rightarrow (1,0)$: $\mathbf{r} = \langle t, 0 \rangle$, integral $= \int_0^1 0 dt = 0$
 C_2 : $(1,0) \rightarrow (0,1)$: $\mathbf{r} = \langle 1-t, t \rangle$, $dx = -dt$, $dy = dt$
 Integral $= \int_0^1 [-(1-t)^2t + (1-t)t^2] dt = \int_0^1 t(1-t)(2t-1) dt$
 $= \int_0^1 (2t^3 - 3t^2 + t) dt = \frac{1}{2} - 1 + \frac{1}{2} = 0$
 C_3 : $(0,1) \rightarrow (0,0)$: $\mathbf{r} = \langle 0, 1-t \rangle$, integral $= 0$
 Total: $0 + 0 + 0 = 0$ ✓

Example 2: Using Green's to Simplify a Circle Integral

Evaluate $\oint_C (e^x + y^3) dx + (x^3 - \sin y) dy$ where C is the unit circle (CCW).

1. **Identify:** $P = e^x + y^3$, $Q = x^3 - \sin y$
2. **Compute:** $Q_x - P_y = 3x^2 - 3y^2$
3. **Convert to polar:** $\iint_D (3x^2 - 3y^2) dA = \int_0^{2\pi} \int_0^1 3r^2(\cos^2 \theta - \sin^2 \theta) \cdot r dr d\theta$
4. **Evaluate:** $= 3 \int_0^1 r^3 dr \cdot \int_0^{2\pi} \cos 2\theta d\theta = 3 \cdot \frac{1}{4} \cdot 0 = \boxed{0}$

2. Area via Green's Theorem

Area Formulas via Green's Theorem

Setting $Q_x - P_y = 1$ in Green's Theorem gives area formulas. After parameterizing C with $\mathbf{r}(t) = \langle x(t), y(t) \rangle$:

$$\text{Area}(D) = \frac{1}{2} \int_a^b \left(x \frac{dy}{dt} - y \frac{dx}{dt} \right) dt$$

Also written in differential notation: $\frac{1}{2} \oint_C (x dy - y dx)$, or equivalently $\oint_C x dy$ or $-\oint_C y dx$.

Example 3: Area of an Ellipse

Find the area enclosed by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ using Green's Theorem.

1. **Parameterize:** $x = a \cos t$, $y = b \sin t$, $t \in [0, 2\pi]$
2. **Compute differentials:** $dx = -a \sin t dt$, $dy = b \cos t dt$
3. **Apply area formula:**

$$A = \frac{1}{2} \oint_C (x dy - y dx) = \frac{1}{2} \int_0^{2\pi} [a \cos t \cdot b \cos t - b \sin t \cdot (-a \sin t)] dt$$

4. **Simplify:** $= \frac{ab}{2} \int_0^{2\pi} (\cos^2 t + \sin^2 t) dt = \frac{ab}{2} \cdot 2\pi = \boxed{\pi ab}$

3. When Green's Theorem Fails

- Green's requires P, Q to have continuous partials on **all** of D
- If $\mathbf{F}$ is undefined inside D (e.g., at the origin), the theorem does **not** apply directly

Example 4: The $1/(x^2 + y^2)$ Counterexample

Compute $\oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$ around the unit circle. Show Green's Theorem does not apply.

1. **Check partials:** $P = \frac{-y}{x^2 + y^2}$, $Q = \frac{x}{x^2 + y^2}$.

$$P_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad Q_x = \frac{y^2 - x^2}{(x^2 + y^2)^2}. \quad \text{So } Q_x - P_y = 0!$$

If Green's applied, the answer would be $\iint_D 0 dA = 0$. But...

2. **Direct computation:** $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$, $\mathbf{r}' = \langle -\sin t, \cos t \rangle$

$$\mathbf{F} \cdot \mathbf{r}' = (-\sin t)(-\sin t) + (\cos t)(\cos t) = 1$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} 1 dt = \boxed{2\pi} \neq 0$$

3. **Why:** $\mathbf{F}$ is undefined at $(0, 0)$, which lies inside C . The domain has a “hole,” so Green's Theorem does not apply.

Practice Problems

Problem 1. Use Green's Theorem to evaluate $\oint_C (2y \, dx + 3x \, dy)$ around the unit square $[0, 1]^2$ (CCW).

Answer: 1

Problem 2. Evaluate $\oint_C (y^2 \, dx + x^2 \, dy)$ where C is the boundary of the region between $y = x$ and $y = x^2$, $0 \leq x \leq 1$ (CCW).

Answer: $\frac{1}{30}$

Problem 3. Use Green's Theorem to find the area enclosed by the astroid $x = \cos^3 t$, $y = \sin^3 t$, $0 \leq t \leq 2\pi$.

Answer: $\frac{3\pi}{8}$

Problem 4. Evaluate $\oint_C (x^2 y \, dx - x y^2 \, dy)$ where C is the circle $x^2 + y^2 = 4$ (CCW).

Answer: $Q_x - P_y = -y^2 - x^2 = -(x^2 + y^2)$; $\iint_D -(x^2 + y^2) \, dA = -\int_0^{2\pi} \int_0^2 r^3 \, dr \, d\theta = -8\pi$

Problem 5. True or false (explain briefly):

(a) If $Q_x - P_y = 0$ everywhere in D , then $\oint_C P dx + Q dy = 0$.

(b) Green's Theorem can be applied to $\mathbf{F} = \langle x/r^2, y/r^2 \rangle$ (where $r = \sqrt{x^2 + y^2}$) on any region not containing the origin.

Answer: (a) True — by Green's Theorem, $\oint 0 dA = 0$, provided P, Q are smooth on all of D .

(b) True — as long as D does not contain the origin, $\mathbf{F}$ is smooth on D and Green's applies.